

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

OCTOBER 2021 TRIMESTER

FINAL ASSESSMENT

**UCCN1223 CYBERSECURITY**

FRIDAY, 10 DECEMBER 2021

TIME : 6.00 PM – 9.00 PM (3 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)  
BACHELOR OF INFORMATION SYSTEMS (HONOURS)  
BUSINESS INFORMATION SYSTEMS  
BACHELOR OF INFORMATION SYSTEMS (HONOURS)  
INFORMATION SYSTEMS ENGINEERING  
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)  
COMPUTER ENGINEERING  
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)  
COMMUNICATIONS AND NETWORKING  
BACHELOR OF INFORMATION SYSTEMS (HONOURS)  
DIGITAL ECONOMY TECHNOLOGY

**Instructions to Candidates:**

REMINDER: You are reminded to read and adhere to the Final Assessment Instructions to Candidates that has been made available through the UTAR Portal before the commencement of Final Assessment (FA). The detailed instructions for this FA are as follows:

**General**

1. This Final Assessment (FA) is an Individual, time restricted assessment which consists of FOUR (4) questions. Each question carries 25 marks.
2. You are required to answer **ALL** questions, and submit the **ANSWER SCRIPT** by **9.00 pm, 10 DECEMBER 2021**.
3. During the period of [**3 hours**] of this FA, the examiner(s) can be reached at
  - (a). [WBLE, Microsoft Teams or Whatsapp, etc.] with Code/Password:[Specific Code/Password created], or
  - (b). Email:[Email Address designated by the Examiner]You may use the above e-platform(s) to check with the examiner(s) if you need any clarification on this FA question paper.
4. You may refer to any books, lecture notes, published materials, online resources etc. Candidates are reminded that **Copy-and-Paste, Consultation, Discussion and Sharing of Answers** with others are **STRICTLY PROHIBITED** in this FA.

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This final assessment paper consists of 4 questions on 6 printed pages.

**UCCN1223 CYBERSECURITY****Answer Script File**

5. The answer script **MUST** be a **Microsoft Word file** [Compulsory for text-based submission to enable Turnitin Plagiarism Check] or opt for PDF and IMAGE file [for non-text/ hand written submission], in A4 size format. **Note: Please keep the file size NOT exceeding 30MB per file.**
6. Please check your Index Number generated by the Division of Examinations, Awards, and Scholarships (DEAS). You **MUST** name your answer script file using the following file name for submission:

**[Course code]\_FA\_[Programme Abbreviation]\_[Your Index Number]**

For example, if you are from the degree programme IB, and your Index Number is A99999CBIBF, then your answer script should be named as:

**UCCN1223\_FA\_IB\_A99999CBIBF.doc**

**Answer Script File Submission**

7. Your answer script file has to be submitted to the following platform(s) before the due time/date.
  - (a). Attach your answer script at WBLE/ Google Form / Microsoft Form (enter the URL link)  
**Note: Use your “1UTAR” email account to access the Google Form. You can edit your submission before the form is closed.**
  - (b). Send your answer script to the course code email  
**Note: For the subject / title of your email, use the file name of your answer script, i.e.,**  
**UCCN1223\_FA\_[Programme Abbreviation]\_[Your Index Number]**
8. Please make sure you submit the correct, complete and final version of your answer script and the same copy of answer script file is to be submitted to the above platform(s). When multiple answer script files are received, the examiner will randomly choose one for marking.

**UCCN1223 CYBERSECURITY****Contents of Answer Script**

9. The **first page** of your submission is the **Final Assessment Cover Page**. You **MUST** use the template given and fill in the following information
  - Your Programme (Abbreviation)
  - Your Index Number
  - Your Name
  - Your Student ID
10. The **second page** of your submission is the **Final Assessment Declaration Statement**. You **MUST** use the template given, and digitally sign on the form to indicate the authenticity of your submitted work is without plagiarism.
11. Each question should be answered starting on a new page.
12. For answer scripts that have text-based answers only, all texts **MUST** be typed and recommended using Times New Roman characters with font size **12**, with proper spacing and alignment, except for the drawings and equations/calculations.
13. For answer scripts that require / contain drawings, equations and calculations with short text descriptions, you can hand-write your answers on a blank paper, and then use the scanner apps in your smartphone (Scanner Document, CamScanner, Adobe Scanner etc.) to take scanned copy, or you can type if necessary and include the scanned copy taken in the Word document, as part of your submitted answers.
14. Please include a page number on **each and every page** of your answer script. Ensure that each page of answer scripts is in sequence prior to online submission.

**WARNING OF PLAGIARISM**

15. Answer scripts shall be uploaded to Turnitin to check for the originality of submitted answers. In the case of suspected plagiarism, the evidence will be submitted to the Examination Disciplinary Committee of the University. Disciplinary action shall be taken against any candidate who is found to have plagiarized in the answer submitted. Hence, candidates are reminded to abide by all University Rules and Regulations and any instructions/guidelines relating to examinations/assessments.

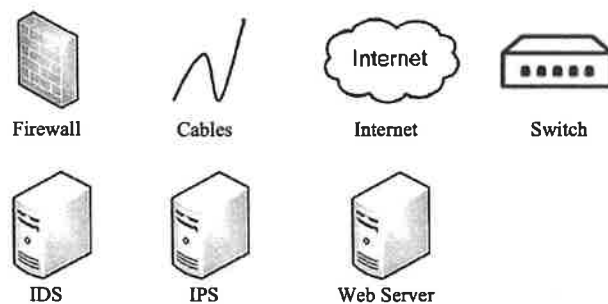
**UCCN1223 CYBERSECURITY**

Q1. MD5 can be used to hash an input data to produce a hash digest.

- (a) MD5 is known to be vulnerable to rainbow table attacks where a password can be lookup from pre-computed hash digests. Suggest **ONE (1)** method to prevent rainbow table attack. (4 marks)
- (b) Explain hash collision;  $H(M)=H(N)$  using an example. (6 marks)
- (c) Suggest **ONE (1)** method to reduce the chances of hash collision. (4 marks)
- (d) Given public key  $P_{PBkey}\{e=17, N=11\}$  and private key  $P_{PRkey}\{d=53, N=11\}$ . Use RSA scheme to encrypt the plaintext 'C=2'. (6 marks)
- (e) Given a public key  $P_{PBkey}\{e=5, N=119\}$ . It is mathematically difficult to find the corresponding private key  $P_{PRkey}\{d, N\}$ . Explain why? (5 marks)

[Total : 25 marks]

- Q2. (a) Intrusion Detection System (IDS) is designed to detect a number of network threats. Identify **THREE (3)** common network threats and briefly explain how they work. (9 marks)
- (b) You are provided with a list of equipment for securing a company's network as shown in Figure 1. Design a secure network using **ALL** of the equipment provided to prevent network threats. You may use the equipment more than once. Briefly explain the reasons for the positioning of the equipment in your network. (10 marks)



**Figure 1**

**UCCN1223 CYBERSECURITY****Q2. (Continued)**

- (c) The goal of a demilitarized zone (DMZ) is to add an extra layer of security to an organization's Local Area Network (LAN). Answer the following questions:
- (i) Suggest a location to place the DMZ in the LAN. (2 marks)
  - (ii) Discuss the disadvantages of dual firewall DMZ architecture. (4 marks)
- [Total : 25 marks]

Q3. (a) **Case study:** AvoTherapy is a specialist hospital for plastic surgery. In Oct 2021, a junior staff named *Billie* leaked some sensitive data of their patients to the darkweb. This resulted in data leakage to public and caused the hospital to suffer a loss of \$1 million.

- (i) Propose **TWO (2)** security measures to improve network security and prevent future data leakage. Explain your answer. (4 marks)
- (ii) Some usernames and passwords of the customers are leaked as a textfile named 'rockyou.txt'. State the Linux command to find out if Taylor's passwords are leaked in this file (assume username is *Taylor*). (4 marks)
- (iii) Suppose that Billie wants to further enumerate the ssh server for more open services after being locked out from ssh. State the nmap command to scan the server, to detect the OS types and other services that are running including those running on hidden ports. The scan should be as stealthy as possible to prevent triggering any alarms. Assume server IP is 10.10.10.10 (4 marks)
- (iv) Billie tried to transfer a file called *secretrecipe.exe* from the *ssh* to her local machine by uploading it as an attachment to Gmail. She later found that Gmail blocked the upload due to security reasons. Suggest **ONE (1)** method for Billie to transfer the executable file from the server to her own machine. (assume *ssh* server IP is 10.10.10.10 and Billie's IP is 10.10.10.1). (4 marks)

- (b) A security policy is a well-defined set of rules which includes information policy, security policy, computer usage policy, and internet usage policy, etc.
- (i) Provide **TWO (2)** function(s) of security policy. (6 marks)
  - (ii) Suggest **ONE (1)** action that can be done by the organization so that the security policy can be properly enforced in the campus network. Explain your answer. (3 marks)
- [Total : 25 marks]

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- Q4. (a) **Case Study 1:** Jennie wants to buy a new personal computer and is browsing online for a good online deal. She sees an advertisement displaying, “25% discount on any Apple Computer!” on a deal-sharing website. She gets excited and clicks on this advertisement. As the page is loading it displays a message saying, “Sorry, our offers expired!!”
- (i) Describe a malware attack that causes the victim to receive this advertisement. (4 marks)
  - (ii) Suggest **TWO (2)** countermeasures against such malware attack. (6 marks)
  - (iii) Suppose that Jennie clicked on the offer and got redirected to *applee.com* instead of *apple.com*. Name and explain the type of cyber attack that Jennie has fallen victim to and suggest **ONE (1)** method to detect such attack. (6 marks)
- (b) **Case Study 2:** Lisa works as a web admin in UTAR. She tendered her resignation to the manager to join TARC. The manager got suspicious that she is stealing trade secrets by joining a rival institution and called for an investigation.
- (i) As a data forensic engineer, what would you do to collect information on this case? (4 marks)
  - (ii) The resignation letter is a pdf file named “sus.pdf”. The manager claims that his PC is infected with virus after opening the file. Suggest **ONE (1)** analysis method you can perform to analyze for malware on the pdf file. (5 marks)
- [Total : 25 marks]